QUBO Formulation: Stereo Matching Disparity Assignment

Stereo Matching Disparity Assignment

Problem Statement. Given a set of pixels $\mathcal{P}$ with $|\mathcal{P}| = P$ and a discrete set of disparity labels $\mathcal{D}$ with $|\mathcal{D}| = D$, the goal is to assign exactly one disparity label to each pixel such that the total stereo energy is minimized. The energy consists of data costs associated with assigning specific disparities to pixels and smoothness penalties incurred when neighboring pixels are assigned different disparities. Let $\mathcal{N} \subseteq \mathcal{P} \times \mathcal{P}$ denote the set of neighboring pixel pairs. For each pixel $p \in \mathcal{P}$ and disparity $d \in \mathcal{D}$, let $C_{p,d} \geq 0$ be the data cost. For each neighbor pair $(p, q) \in \mathcal{N}$, let $V_{p,q} \geq 0$ be the smoothness weight. The objective is to find a disparity assignment that minimizes the sum of selected data costs and the sum of smoothness penalties between neighboring pixels.

Decision Variables. We introduce binary variables $x_{p,d} \in \{0, 1\}$ for all $p \in \mathcal{P}$ and $d \in \mathcal{D}$. The variable $x_{p,d}$ indicates whether pixel p is assigned disparity d, such that $x_{p,d} = 1$ if pixel p is assigned disparity d, and $x_{p,d} = 0$ otherwise.

QUBO Derivation. We derive the Quadratic Unconstrained Binary Optimization (QUBO) formulation by converting the constrained minimization problem into an unconstrained energy minimization problem using penalty terms.

Step 1: Original Objective. The stereo energy to be minimized is given by:

$$E(x) = \sum_{p\in\mathcal{P}} \sum_{d\in\mathcal{D}} C_{p,d} x_{p,d} + \sum_{(p,q)\in\mathcal{N}} V_{p,q} \left(1 - \sum_{d\in\mathcal{D}} x_{p,d} x_{q,d} \right). \tag{1}$$

Expanding the smoothness term yields:

$$E(x) = \sum_{p\in\mathcal{P}} \sum_{d\in\mathcal{D}} C_{p,d} x_{p,d} + \sum_{(p,q)\in\mathcal{N}} V_{p,q} - \sum_{(p,q)\in\mathcal{N}} \sum_{d\in\mathcal{D}} V_{p,q} x_{p,d} x_{q,d}. \tag{2}$$

Step 2: Constraints. Each pixel must be assigned exactly one disparity. This is enforced by the constraints:

$$\sum_{d\in\mathcal{D}} x_{p,d} = 1, \quad \forall p \in \mathcal{P}. \tag{3}$$

Step 3: Penalty Terms. We convert each constraint into a penalty term using a weight $\lambda > 0$. For each pixel p, the penalty is:

$$H_{\text{pen},p}(x) = \lambda \left(\sum_{d \in \mathcal{D}} x_{p,d} - 1 \right)^2 . \tag{4}$$

Expanding the square and using the idempotent property $x_{p,d}^2 = x_{p,d}$ for binary variables:

$$\begin{aligned} H_{\text{pen},p}(x) &= \lambda \left(\sum_{d \in \mathcal{D}} x_{p,d}^2 + \sum_{d \in \mathcal{D}} \sum_{d' \in \mathcal{D}, d \neq d'} x_{p,d} x_{p,d'} - 2 \sum_{d \in \mathcal{D}} x_{p,d} + 1 \right) & (5) \\ &= \lambda \left(\sum_{d \in \mathcal{D}} x_{p,d} + \sum_{d \neq d'} x_{p,d} x_{p,d'} - 2 \sum_{d \in \mathcal{D}} x_{p,d} + 1 \right) & (6) \\ &= \lambda \left(\sum_{d \neq d'} x_{p,d} x_{p,d'} - \sum_{d \in \mathcal{D}} x_{p,d} + 1 \right) . & (7) \end{aligned}$$

The total penalty is $H_{\text{pen}}(x) = \sum_{p \in \mathcal{P}} H_{\text{pen},p}(x)$.

Step 4: Combined Hamiltonian. The QUBO Hamiltonian $H(x) = E(x) + H_{\text{pen}}(x)$ is:

$$\begin{aligned} H(x) = &\sum_{p \in \mathcal{P}} \sum_{d \in \mathcal{D}} C_{p,d} x_{p,d} + \sum_{(p,q) \in \mathcal{N}} V_{p,q} - \sum_{(p,q) \in \mathcal{N}} \sum_{d \in \mathcal{D}} V_{p,q} x_{p,d} x_{q,d} \\ &+ \sum_{p \in \mathcal{P}} \lambda \sum_{d \neq d'} x_{p,d} x_{p,d'} - \sum_{p \in \mathcal{P}} \lambda \sum_{d \in \mathcal{D}} x_{p,d} + \sum_{p \in \mathcal{P}} \lambda . \end{aligned} \tag{8}$$

Grouping linear, quadratic, and constant terms:

$$\begin{aligned} H(x) = &\sum_{p \in \mathcal{P}} \sum_{d \in \mathcal{D}} (C_{p,d} - \lambda) x_{p,d} - \sum_{(p,q) \in \mathcal{N}} \sum_{d \in \mathcal{D}} V_{p,q} x_{p,d} x_{q,d} \\ &+ \sum_{p \in \mathcal{P}} \lambda \sum_{d \neq d'} x_{p,d} x_{p,d'} + \sum_{(p,q) \in \mathcal{N}} V_{p,q} + P\lambda . \end{aligned} \tag{9}$$

Step 5: Q Matrix and Offset. The Hamiltonian is of the form $H(x) = x^\top Q x + c$. Mapping variables to indices $i = pD + d$, the Q matrix entries and offset are:

$$\begin{aligned} Q_{(p,d),(p,d)} &= C_{p,d} - \lambda, & \forall p \in \mathcal{P}, d \in \mathcal{D}, \quad (10) \\ Q_{(p,d),(p,d')} &= 2\lambda, & \forall p \in \mathcal{P}, d, d' \in \mathcal{D}, d \neq d', \quad (11) \\ Q_{(p,d),(q,d)} &= -V_{p,q}, & \forall (p,q) \in \mathcal{N}, d \in \mathcal{D}, \quad (12) \\ Q_{i,j} &= 0, & \text{otherwise}, \quad (13) \end{aligned}$$

with offset

$$c = \sum_{(p,q) \in \mathcal{N}} V_{p,q} + P\lambda . \tag{14}$$

Penalty Weight Justification. To ensure that the global minimum of the QUBO corresponds to a feasible solution, the penalty weight λ must be chosen sufficiently large. Specifically, λ must exceed the maximum possible reduction in the objective function achievable by violating the constraint for any single pixel. The maximum data cost associated with pixel p is $\sum_{d \in \mathcal{D}} C_{p,d}$, and the maximum smoothness penalty involving pixel p is $\sum_{q \in \mathcal{N}(p)} V_{p,q}$. Thus, a sufficient condition is:

$$\lambda > \max_{p \in \mathcal{P}} \left(\sum_{d \in \mathcal{D}} C_{p,d} + \sum_{q \in \mathcal{N}(p)} V_{p,q} \right). \tag{15}$$

This guarantees that any violation of the constraint incurs a penalty strictly greater than any potential gain in the objective function.

Correctness Argument. Minimizing $H(x)$ over $\{0,1\}^{P \times D}$ recovers the optimal disparity assignment. For any feasible solution, the penalty terms vanish, and $H(x) = E(x)$. For any infeasible solution, at least one constraint is violated, incurring a penalty of at least λ. By the choice of λ, the energy of any infeasible solution exceeds the energy of the best feasible solution. Therefore, the global minimum of $H(x)$ must be feasible and corresponds to the assignment minimizing the original stereo energy.

Illustrative Example. Consider a small instance with $P = 2$ pixels ($\mathcal{P} = \{0,1\}$), $D = 3$ disparities ($\mathcal{D} = \{0,1,2\}$), and a single neighbor pair $\mathcal{N} = \{(0,1)\}$ with weight $V_{0,1}$. The variables are indexed as $i = 3p + d$, yielding $x_{0,0}, x_{0,1}, x_{0,2}, x_{1,0}, x_{1,1}, x_{1,2}$ at indices $0, \ldots, 5$.

The objective is:

$$E(x) = \sum_{p=0}^{1} \sum_{d=0}^{2} C_{p,d} x_{p,d} + V_{0,1} \left(1 - \sum_{d=0}^{2} x_{0,d} x_{1,d} \right). \tag{16}$$

The penalty terms for $p = 0$ and $p = 1$ are:

$$H_{\text{pen}}(x) = \lambda \sum_{p=0}^{1} \left(\sum_{d \neq d'} x_{p,d} x_{p,d'} - \sum_{d=0}^{2} x_{p,d} + 1 \right). \tag{17}$$

The resulting Hamiltonian $H(x)$ has the following Q matrix entries:

$$Q_{0,0} = C_{0,0} - \lambda, \qquad Q_{0,1} = 2\lambda, \qquad Q_{0,3} = -V_{0,1}, \tag{18}$$
$$Q_{1,1} = C_{0,1} - \lambda, \qquad Q_{1,2} = 2\lambda, \qquad Q_{1,4} = -V_{0,1}, \tag{19}$$
$$Q_{2,2} = C_{0,2} - \lambda, \qquad Q_{2,5} = -V_{0,1}, \qquad Q_{3,3} = C_{1,0} - \lambda, \tag{20}$$
$$Q_{3,4} = 2\lambda, \qquad Q_{3,5} = 2\lambda, \qquad Q_{4,4} = C_{1,1} - \lambda, \tag{21}$$
$$Q_{4,5} = 2\lambda, \qquad Q_{5,5} = C_{1,2} - \lambda. \tag{22}$$

All other off-diagonal entries are zero. The offset is $c = V_{0,1} + 2\lambda$. Minimizing $H(x)$ yields the optimal disparity assignment for this instance.